# Lecture 7

**Normal Subgroups:** A subgroup $H$ of a group $G$ is called a normal subgroup of $G$ if it satisfies

$$aH = Ha \quad \forall\, a \in G.$$

We write $H \trianglelefteq G$

**Example:** Any subgroup $H$ of an abelian group is normal.

$aH = \{ah \mid h \in H\}$

$Ha = \{ha \mid h \in H\}$

Since the group is abelian $ah = ha \;\; \forall\, a \in G,\, h \in H.$

Hence $aH = Ha \;\; \forall\, a \in G.$

Consider the group $S_3 = \{e, (12), (13), (23), (123), (132)\}$

Take $H = \{e, (123), (132)\}$

Then $H \leq S_3.$

One can also check $H \trianglelefteq G$ i.e. $H$ is normal in $G$.

Any group $G$ has two trivial subgroups, namely $\{e\}$ & $G$.

Both these subgroups are normal in $G$. i.e.

$$\{e\} \trianglelefteq G \quad \& \quad G \trianglelefteq G.$$

**Simple group:** A group $G \neq \{e\}$ is called a simple group if the only normal subgroups of $G$ are $\{e\}$ and $G$; i.e. $G$ has no non-trivial normal subgroup.

**Theorem:** A subgroup $H$ of a group $G$ is a normal subgroup of $G$ if and only if

$$ghg^{-1} \in H \quad \forall g \in G \;\&\; h \in H.$$

**Proof:** Let $H$ is a normal subgroup of $G$.

Let $g \in G$ & $h \in H$ be arbitrary elements.

Need to prove: $ghg^{-1} \in H$.

Since $H$ is normal in $G$, we have

$$gH = Hg.$$

So $gh \in gH = Hg. = \{h'g \mid h' \in H\}$

$\Rightarrow \exists h_1 \in H$ such that $gh = h_1 g$

$\Rightarrow ghg^{-1} = h_1 \in H$. [By multiplying $g^{-1}$ on the right]

**Conversely:** Let $ghg^{-1} \in H \quad \forall g \in G$ & $h \in H$.

Need to prove: $gH = Hg \quad \forall g \in G$.

So it is enough to prove

$$gH \subseteq Hg \quad \& \quad Hg \subseteq gH.$$

Let us first prove that $gH \subseteq Hg$

Let $gh \in gH$. where $h \in H$.

Since $ghg^{-1} \in H$, $ghg^{-1} = h_1$ for some $h_1 \in H$.

$\Rightarrow gh = h_1 g$ [By multiplying $g$ on the right on both sides]

$\Rightarrow gh = h_1 g \in Hg. \quad \Rightarrow gH \subseteq Hg.$

Similarly one can prove that

$$Hg \subseteq gH.$$

Hence $gH = Hg$. $\Rightarrow H \trianglelefteq G$.

<u>Remark:</u> Let $H$ be a normal subgroup of $G$.

So $gH = Hg \quad \forall\, g \in G$.

$gH$ & $Hg$ are two subsets of $G$ & as a subset they are same.

$gh \in gH = Hg$ but <u>this doesn't mean</u> that

$$gh = hg$$

All it means is that $gh \in Hg$ i.e.

$gh = h_1 g$ for some $h_1 \in H$.

$h$ & $h_1$ <u>need not be same</u>.

<u>Lemma:</u> Let $G$ be a group & $H \leq G$. Then for any $g \in G$, $|gH| = |H| = |Hg|$.

<u>Proof:</u> To prove we will define a bijection between $gH$ and $H$.

Define $f : H \longrightarrow gH$

$$f(h) = gh.$$

Claim $f$ is a bijection.

Let us first prove that $f$ is one-one.

Let $f(h_1) = f(h_2)$ for $h_1, h_2 \in H$

$\Rightarrow gh_1 = gh_2$

$\Rightarrow g^{-1}(gh_1) = g^{-1}(gh_2)$

$\Rightarrow ((g^{-1}g)h_1) = ((g^{-1}g)h_2)$

$\Rightarrow h_1 = h_2$. $\Rightarrow f$ is one-one.

~~Let to~~ Now let us prove that $f$ is onto.

Let $x \in gH$

then $x = gh$ for some $h \in H$.

$\Rightarrow f(h) = x$.

$\Rightarrow f$ is onto.

So $f$ is a bijection between $H$ & $gH$.

Hence $|H| = |gH|$.

Similarly one can prove that the map

$$f' : H \longrightarrow Hg$$
$$h \longrightarrow hg$$

is a bijection.

Combining these we get

$$|Hg| = |H| = |gH| \quad \forall g \in G.$$

**Order of a group:** The number of elements in the group $G$ is called the order of the group and is denoted by $|G|$.

A group of finite order is called a finite group.

**Lagrange's Theorem:** Let $G$ be a finite group & $H \leq G$. Then $|H| \,\big|\, |G|$.

**Proof:** Let $g_1H, g_2H, \ldots, g_kH$ be all the distinct left cosets of $H$ in $G$.

Since these left cosets are equivalence classes under the left congruence relation modulo $H$ ($\equiv_L$), they form a partition of $G$. i.e.

$$G = g_1H \amalg g_2H \amalg \cdots \amalg g_kH \quad \text{(disjoint union)}$$

So $$|G| = \sum_{i=1}^{k} |g_iH|$$

but $|g_iH| = |H| \quad \forall\, g_i \in G$.

$$\Rightarrow |G| = k\,|H| \quad \Rightarrow |H| \,\big|\, |G|.$$

So $k = \dfrac{|G|}{|H|}$ is the number of distinct left cosets of $H$ in $G$.

By considering the right cosets instead of left cosets in the proof, one can also prove that $k = \dfrac{|G|}{|H|}$ is the number of distinct right cosets of $H$ in $G$.

**Remark:** The number of distinct left and right cosets of a subgroup $H$ in a group $G$ is equal.

**Index of a subgroup:** Let $G$ be a group & $H \leq G$. Then the number of distinct left (or right) cosets of $H$ in $G$ is called the index of $H$ in $G$ & is denoted by $[G:H]$.

**Eg:** Let $G = S_3 = \{e, (12), (13), (23), (123), (132)\}$.

$H_1 = \{e, (12)\}$ $\quad H_1 \leq S_3$.

$$[S_3 : H_1] = \frac{|S_3|}{|H_1|} = \frac{6}{2} = 3$$

$H_2 = \{e, (123), (132)\}$ $\quad H_2 \leq S_3$.

$$[S_3 : H_2] = \frac{|S_3|}{|H_2|} = \frac{6}{3} = 2.$$

$G = \mathbb{Z}_8 = \{[0], [1], \dots, [7]\}$. under addition.

$H_1 = \{[0], [4]\}$ $\quad H_1 \leq \mathbb{Z}_8$.

$$[G : H_1] = \frac{|\mathbb{Z}_8|}{|H_1|} = \frac{8}{2} = 4.$$

$H_2 = \{[0], [2], [4], [6]\}$ $\quad H_2 \leq \mathbb{Z}_8$.

$$[\mathbb{Z}_8 : H_2] = \frac{|\mathbb{Z}_8|}{|H_2|} = \frac{8}{4} = 2.$$

Converse of Lagrange's theorem doesn't hold in genereal.

We will prove that the converse of Lagrange's theorem holds for finite abelian group. We will also discuss an example of a finite non-abelian group for which the converse of Lagrange's theorem doesn't hold.

Order of an element: Let $G$ be a group & let $g \in G$. If there exists a positive integer $n$ such that $g^n = 1$ (identity of $G$), then the element $g$ is called an element of finite order. And the smallest positive integer $n$ for which $g^n = 1$ is called the order of the element $g$.

If no such positive integer $n$ exists, then $g$ is called an element of infinite order.

Example: ① $G = (\mathbb{Z}_8, +)$.

Let $[4] \in G$.

$[4] + [4] = [8] = [0]$ (identity of $\mathbb{Z}_8$)

$\Rightarrow$ order of $[4]$ is 2.

② $G = S_3$. $(12) \in S_3$.

$(12)(12) = e \in S_3$.

$\Rightarrow$ order of $(12)$ is 2.

$(123) \in S_3$

$(123)(123) = (132)$

$(123)(123)(123) = e$.

$\Rightarrow$ order of $(123)$ is 3.

**Note:** If $G$ is an additive group & $g \in G$, then the smallest positive integer $n$ such that $ng = 0$ (identity) is the order of the element $g$.

And the element $g$ is called of infinite order if $ng \neq 0$ ~~for~~ $\forall n \in \mathbb{N}$.

Order of an element $g$ is denoted by $|g|$.

**Eg:** $G = (\mathbb{Z}, +)$.

$1 \in \mathbb{Z}$. $\quad$ $1$ is an element of infinite order.

as $\quad n.1 = n \neq 0 \quad \forall n \in \mathbb{N}$.

**Theorem:** In a finite group, each element is of finite order.

**Proof:** Let $G$ be ~~the~~ a finite group with identity $1$.

Let $g \in G$.

Then $\quad g^2 = g \cdot g \in G$.

$g^3 = g \cdot g \cdot g \in G$

$\vdots$

$g^n = g \cdot g \cdots g$ ($n$ times) $\in G$

So $\{g, g^2, g^3, \ldots, g^n, \ldots\} \subseteq G$.

Since $G$ is finite, $\exists$ positive integers $i$ & $j$ with $i > j$ such that

$$g^i = g^j$$

$\Rightarrow (g^i)(g^j)^{-1} = (g^j)(g^j)^{-1}$ 　　$\left[\because (g^j)^{-1} = g^{-j}\right.$

$\Rightarrow (g^i)(g^{-1})^j = 1$

$\Rightarrow g^{i-j} = 1.$

Since $i > j$, $i-j$ is a positive integer.

$\Rightarrow |g|$ is finite.

<u>Cyclic group</u>: A group $G$ is called a cyclic group if $\exists\, g \in G$ such that

$$G = \{ g^n \mid n \in \mathbb{Z} \}.$$

The element $g$ is called a generator of $G$.

Note: If $G$ is an additive group then ~~g~~ $G$ is called cyclic if $\exists\, g \in G$ such that

$$G = \{ ng \mid n \in \mathbb{Z} \}.$$

And the element $g$ is called a generator.

<u>Example</u>: (1) $(\mathbb{Z}, +)$.

$$\mathbb{Z} = \{ n.1 \mid n \in \mathbb{Z} \}.$$
$$= \{ n.(-1) \mid n \in \mathbb{Z} \}.$$

Hence $\mathbb{Z}$ is cyclic & $1, -1$ are generators of $\mathbb{Z}$.

(2) $(\mathbb{Z}_4, \oplus)$.

$\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$.

$\mathbb{Z}_4 = \{n \cdot [1] \mid n \in \mathbb{Z}\}$

So $\mathbb{Z}_4$ is cyclic & $[1]$ is a generator of $\mathbb{Z}_4$.

Check $[3]$ is also a generator of $\mathbb{Z}_4$.

Theorem: A group $G$ of order $n$ is cyclic if and only if it has an element of order $n$.

Proof: Let $g \in G$ be an element of order $n$.

then $1, g, g^2, \ldots, g^{n-1}$ are distinct elements of $G$.

Since $|G| = n$ we have

$$G = \{1, g, g^2, \ldots, g^{n-1}\}$$
$$\subseteq \langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}.$$

~~So G~~

We also have $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\} \subseteq G$ as $G$ is closed.

$$\Rightarrow G = \langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}.$$

$\Rightarrow G$ is cyclic.

Conversely, let $G$ be a cyclic group of order $n$.

i.e. $\exists\ g \in G$ such that

$$G = \langle g \rangle = \{ g^m \mid m \in \mathbb{Z} \}.$$

We need to prove $G$ has an element of order $n$.

We claim that $|g| = n$.